

Experiment 3

Implementation of Triple Data Encryption Standard (3DES)

Aim

To implement the Triple Data Encryption Standard (3DES) algorithm for encrypting and decrypting plaintext using three secret keys.

Description

Triple Data Encryption Standard (3DES) is an enhanced version of the DES algorithm that applies the encryption process three times using three different keys. The plaintext is encrypted with the first key, encrypted again with the second key, and finally encrypted with the third key. During decryption, the reverse process is performed using the keys in reverse order. Applying three encryption operations provides stronger security than a single DES operation.

Algorithm

1. Start the program.
2. Read the plaintext from the user.
3. Read three secret keys.
4. Encrypt the plaintext using the first key.
5. Encrypt the result using the second key.
6. Encrypt the result using the third key.
7. Display the encrypted text.
8. Decrypt the encrypted text using the third key.
9. Decrypt the result using the second key.
10. Decrypt the result using the first key.
11. Display the original plaintext.
12. Stop the program.

Code:

```
plaintext = input("Enter Plain Text: ").upper()
key1 = int(input("Enter Key 1: "))
key2 = int(input("Enter Key 2: "))
key3 = int(input("Enter Key 3: "))
encrypted = ""
```

```

for ch in plaintext:
    if ch.isalpha():
        x = (ord(ch) - 65 + key1) % 26
        y = (x + key2) % 26
        z = (y + key3) % 26
        encrypted += chr(z + 65)
    else:
        encrypted += ch
print("Encrypted Text:", encrypted)

decrypted = ""
for ch in encrypted:
    if ch.isalpha():
        x = (ord(ch) - 65 - key3) % 26
        y = (x - key2) % 26
        z = (y - key1) % 26
        decrypted += chr(z + 65)
    else:
        decrypted += ch
print("Decrypted Text:", decrypted)

```

Output:

s

```

IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/abhid/AppData/Local/Programs/Python/Python314/crypto.py ==
Enter Plain Text: Hello can we meet tommorow
Enter Key 1: 12
Enter Key 2: 24
Enter Key 3: 17
Encrypted Text: IFMMP DBO XF NFFU UPNNPSPX
Decrypted Text: HELLO CAN WE MEET TOMMOROW
>>>

```